INDEX · 11 CHAPTERS · 132 FILES

C Practice Programs

c1hello.c → C11Q10.c

WHAT'S IN THE ZIP

132 files across 11 chapters

118

C source files

Mostly one .c file per question

11

Chapters

From Hello World to dynamic memory

14

Other files

C3Q1.exe and text data files

How the files are named

`C4Q7.c = Chapter 4, Question 7 · Qh = homework`

Chapter map

Chapter	Topic	Files
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CHAPTER 1 · 5 FILES

Input, output and arithmetic

File	What it does
c1hello.c	Hello World
C1Q1.c	Sum of two integers
C1Q2.c	Area of a square

File	What it does
C1Q3.c	Circumference of a circle
C1Q4.c	Float sum, product, expression

CHAPTER 2 · 8 FILES

Operators and expressions

File	What it does
C2Q1.c	Float truncated into an int
C2Q2.c	Operator precedence examples
C2Q3.c	Even check with %
C2Q4.c	Odd/even test, prints 1 or 0

File	What it does
C2Q5.c	Is x a two-digit number
C2Qh1.c	Average of x, y, z
C2Qh2.c	Is a character a digit 1 to 9
C2Qh3.c	Digit check, second copy

CHAPTER 3 · 8 FILES

Conditions: if, switch, ternary

File	What it does
C3Q1.c	Age check with if / else-if
C3Q1.exe	Compiled build of C3Q1
C3Q2.c	Day name via switch (1 to 7)
C3Q3.c	Day name from a letter

File	What it does
C3Q4.c	Pass/fail, grade, percentage
C3Q5.c	The x = 3 inside if trap
C3Qh1.c	Split a 3-digit number
C3Qh2.c	Natural number check, ternary

CHAPTER 4 · 13 FILES

Loops: for, while, do-while

File	What it does
C4Q1.c	Hello World N times
C4Q2.c	Count 0 to 10 with for
C4Q3.c	do-while counting up to N
C4Q4.c	Sum of the first N numbers
C4Q5.c	Multiplication table of n
C4Q6.c	break when i reaches 5
C4Q7.c	Read until an odd number

File	What it does
C4Q8.c	Read until a multiple of 7
C4Q9.c	continue: skip i = 6
C4Q10.c	Step from 5 to 50 by 2
C4Q11.c	Factorial of n
C4Q12.c	Reverse table of n
C4Q13.c	Sum of 5 to 50, with printout

CHAPTER 5 · 19 FILES · PART 1 OF 2

Functions: the basics

File	What it does
C5Q1.c	Declare, call, define printHello
C5Q2.c	Two void functions
C5Q3.c	Greeting chosen by letter
C5Q4.c	printTable with a parameter
C5Q5.c	calculatePrice with a float

File	What it does
C5Q6.c	Square of n using pow()
C5Q7.c	Area calculator menu
C5Q8.c	printHello with a count
C5Q9.c	Sum of the first n numbers
C5Q10.c	Factorial function

CHAPTER 5 · 19 FILES · PART 2 OF 2

Functions: recursion and extras

File	What it does
C5Q11.c	Celsius to Fahrenheit
C5Q12.c	Average of three subjects
C5Q13.c	nth Fibonacci term
C5Q14.c	Recursive Fibonacci, traced
C5Q15.c	main() calling itself

File	What it does
C5Qh1.c	Digit count and digit sum
C5Qh2.c	Square root function
C5Qh3.c	Weather message by temperature
C5Qh4.c	my_pow(base, exp) with checks

CHAPTER 6 · 10 FILES

Pointers

File	What it does
C6Q1.c	Pointer to age, addresses
C6Q2.c	Read and write through *ptr
C6Q3.c	Pointer to a pointer
C6Q4.c	Call by value vs reference
C6Q5.c	swap() with pointers

File	What it does
C6Q6.c	Address of a parameter
C6Q7.c	Sum, product, average via pointers
C6Qh1.c	Greater of two numbers
C6Qh2.c	Print an array in reverse
C6Qh3.c	Walk a string with a pointer

CHAPTER 7 · 11 FILES

Arrays and 2D arrays

File	What it does
C7Q1.c	Marks array: phy, chem, math
C7Q2.c	Item prices and final costs
C7Q3.c	Pointer difference, comparison
C7Q4.c	Read and print 5 values
C7Q5.c	Pass an array to a function
C7Q6.c	2D array marks[2][3]

File	What it does
C7Q7.c	Count odd numbers in an array
C7Q8.c	Reverse an array
C7Q9.c	Fibonacci stored in an array
C7Q10.c	Tables of 2 and 3 in 2D array
C7Q11.c	Same tables, using functions

CHAPTER 8 · 13 FILES

Strings

File	What it does
C8Q1.c	Char array declarations
C8Q2.c	Print first and last name
C8Q3.c	Read names with scanf %s
C8Q4.c	fgets and puts
C8Q5.c	Length of a string
C8Q6.c	Copy with strcpy
C8Q7.c	Compare with strcmp

File	What it does
C8Q8.c	Read a line char by char
C8Q9(i).c	Add salt to a password
C8Q9(ii).c	Salting with a custom function
C8Q10.c	Slice a string, index n to m
C8Q11.c	Count the vowels
C8Q12.c	Find a character with strchr

CHAPTER 9 · 11 FILES

Structures

File	What it does
C9Q1.c	Student struct basics
C9Q2.c	Array of 3 students, preset
C9Q3.c	Read 3 students from input
C9Q4.c	Struct pointer: (*ptr) and ->
C9Q5.c	Pass a struct to a function
C9Q6.c	Address records, 5 people

File	What it does
C9Q7.c	Add two vector structs
C9Q8.c	Complex number via pointer
C9Q9.c	BankAccount with typedef
C9Qh1.c	Staff and student CGPA tracker
C9Qh2.c	Semester-wise CGPA tracker

CHAPTER 10 · 9 FILES

File handling

File	What it does
C10Q1.c	Read a file until EOF
C10Q2i.c	Read 5 characters, fscanf
C10Q2ii.c	fgetc 5 characters, append text
C10Q3.c	Write a sentence to a file
C10Q4.c	Print a whole file, fgetc loop

File	What it does
C10Q5.c	Read 5 integers from a file
C10Q6.c	Append student details
C10Q7.c	Write odd numbers up to n
C10Q8.c	Replace two numbers by their sum

Each program reads or writes a matching .txt file of the same name (C10Q3.txt, C10Q8.txt and so on).

CHAPTER 11 · 10 FILES

Dynamic memory allocation

File	What it does
C11Q1.c	sizeof int, float, char
C11Q2.c	malloc 5 ints and print
C11Q3.c	One malloc: 4 ints, 4 chars
C11Q4.c	calloc: int, chars, float
C11Q5.c	calloc n ints, read, print

File	What it does
C11Q6.c	calloc, free, calloc again
C11Q7.c	Read n ints, then free
C11Q8.c	realloc from 5 to 8 ints
C11Q9.c	calloc 5 ints, user input
C11Q10.c	realloc 5 to 6 ints

OUTSIDE THE CHAPTERS · 15 FILES

Extra programs and data files

File	What it is
zzzz.c	Campus system: student, teacher, staff menu (296 lines)
test.c	Staff details input with fgets and a getchar pause
student / teacher / staff_database.txt	Saved profile records (3 files)
C10Q2i.txt to C10Q8.txt	Data files for Chapter 10 (8 files, plus C10Q6 with no extension)
Test.txt	A single word, APPLE